

..... WESTERN UNIVERSITY
Semester End Examination-2078

Programming Fundamentals and C Programming (CSIT.115)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – A

Attempt all questions (Very Short Questions)

| 8×3 = 24 |

1. Write down characteristics of algorithm.
2. Define flow chart.
3. What is program documentation? Why is it important?
4. Describe application of pointer in brief.
5. What are the key features of c programming language?
6. Describe the difference between = and == operator in C programming?
7. Describe the header file and its usage in C programming?
8. Describe the general form of user defined function in C?

Group – B

Attempt any five questions. (Short Questions)

| 5×8 = 40 |

9. Describe the structure of c program.
10. Explain various types of operators used in C.
11. Write a program using recursive function to find factorial of given number.
12. Define string manipulation function. Describe any three string manipulation function with examples.
13. Write an algorithm and C program to find smallest among three numbers.
14. Explain ftell() and fseek() function. Write a C language program to display the content of file on screen.

Group – C

Attempt any three questions. (Long Questions)

| 3×12 = 36 |

15. Write c program to display multiplication of two matrices.
16. Define sorting. Write c program for insertion sort.
17. Differentiate between structure and union. Write a program to enter record of student (marks, name, address) and display them in ascending order according to marks using structure.
18. Write c program and draw flowchart to check whether entered number is palindrome or not.

FAR WESTERN UNIVERSITY
Semester End Examination-2078
Electronic Principles (CSIT.114)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – A

Attempt all questions (Very Short Questions)

8×3 = 24

1. What is ideal constant current source and ideal constant voltage source?
2. What is the biasing rule for proper functioning of a transistor?
3. What are alpha (α) and beta (β) of a transistor? Write the relation between α and β .
4. What do you understand by JFET and MOSFET?
5. What do you mean d.c. load line? Explain
6. Write your understanding about Barkhausen criteria.
7. Distinguish between regulated and unregulated power supply.
8. Write down the differences between photodiode and LED.

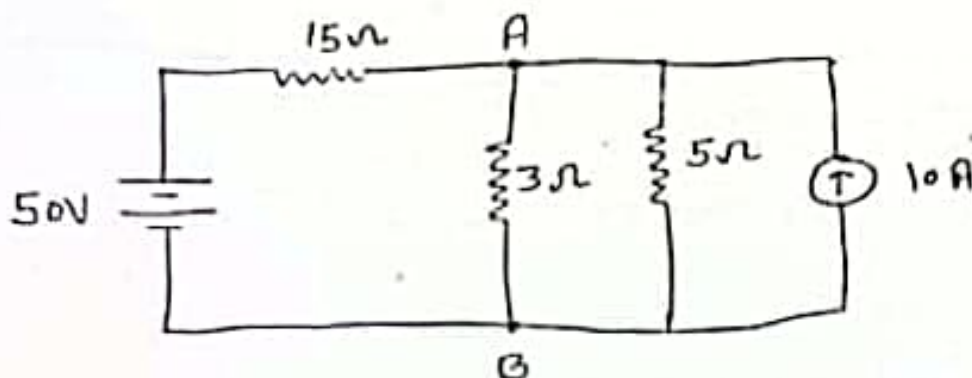
Group – B

Attempt any five questions. (Short Questions)

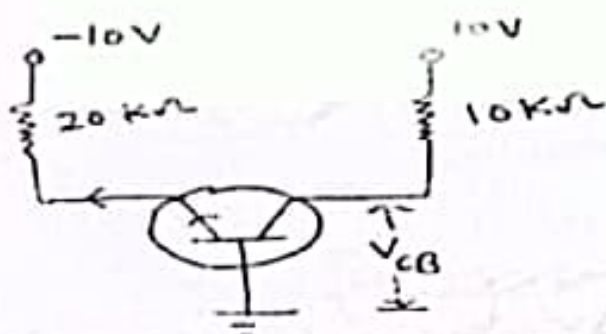
5×8 = 40

9. What is Zener diode? How the Zener diode can be used as voltage regulator?
10. How transistor can be used as an amplifier and as a switch.
11. Define oscillator? Explain the working of Colpitts oscillator.
12. Discuss the working of common emitter (CE) amplifier by drawing its circuit diagram. Also derive the expression for voltage gain.
13. Plot and discuss the JFET drain characteristics with $V_{GS} = 0$. Also discuss transfer characteristics of JFET.
14. Write short notes on following topics:
 - a. Series voltage regulator
 - b. Shunt voltage regulator
 - c. Load regulation
 - d. Line Regulation

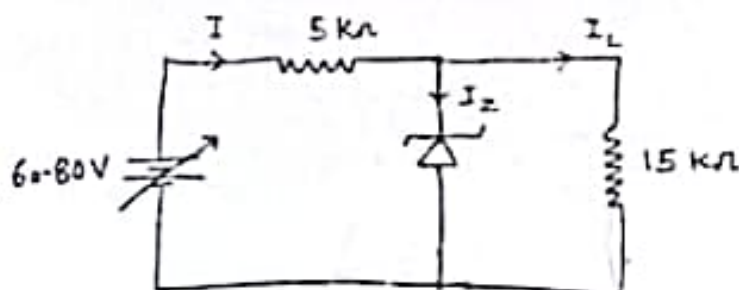
15. a. State and explain the Thevenin's theorem with the help of circuit diagram. [6]
 b. By using the Norton's theorem find the current passing through $3\ \Omega$. [6]



16. a. What is operational amplifier? Explain with the help of circuit diagram the working of inverting operational amplifier. Also deduce the voltage gain for inverting amplifier. [2]
 b. Show that voltage gain of voltage follower amplifier is unit. [4]
 17. a. Draw the circuit diagram for Common Emitter (CE) configuration of a transistor. Also describe the input and output characteristics of CE-transistor. [6]
 b. In the CB circuit given below find the value of V_{CB} . Neglect junction voltage V_{EB} . [6]



18. a. Describe the construction of DE MOSFET. Discuss the characteristics of a DE MOSFET. [6]
 b. Using the ideal Zener diode approximation, find the maximum and minimum currents through the Zener diode. ($V_Z = 30V$) [6]



FAR WESTERN UNIVERSITY
Semester End Examination-2078
Calculus and Analytical Geometry (CSIT.113)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Full Marks: 100

Time: 3hrs.

Attempt all questions (Very Short Questions) 8x3 = 24

- With example define the bounded and unbounded sequences. Identify whether the sequence $\langle u_n \rangle = \frac{n+1}{n}$ is bounded or not.
- What is meant by the limit point of sequence $\langle u_n \rangle$. Show that 0 and 2 are the limit points of sequence defined by $\langle u_n \rangle = 1 + (-1)^n$.
- State the criteria for a series $\sum u_n$ to be convergent. Determine whether the series $\sum u_n = 2^n$ is convergent or non-convergent.
- What do you mean by non-convergent series. Show that the series $\sum \frac{n^2-1}{n^2+1}$ is not convergent.
- State Euler's theorem for homogeneous function and verify it for function $U = \frac{x+y}{x-y}$.
- Test the convergence of the integral $\int_0^2 \frac{dx}{(1-x)^2}$.
- Discuss the motion of a particle whose position $P(x,y)$ at time t is given by $x = a \cos t, y = b \sin t, 0 \leq t \leq 2\pi$.
- Using double integral find area bounded by lines $x=0, y=2x$ and $y=4$.

Group - B

Attempt any five questions. (Short Questions) 5x8=40

- Prove that the series $\sum \frac{1}{n^p}$ converges or diverges according as $p > 1$ or $p \leq 1$. Also test the convergence of series $1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$
- State Leibnitz theorem for n^{th} derivative. If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$ show that $(1-x^2)y_{n+2} - (2n+3)xy_{n+1} - (n+1)^2 y_n = 0$

P.T.O.

11. Obtain the reduction formula for $\int \operatorname{cosec}^n x \, dx$ and evaluate the integral $\int \operatorname{cosec}^6 x \, dx$.
12. Find the vector and parametric equation of the line passing through the points (1,2,3) and (4,6,8). Also find the symmetric equation of same line.
13. Sketch the region of integration for the integral $\iint_R (4x+2) \, dy \, dx$ where $R: 0 \leq x \leq 2, x^2 \leq y \leq 2x$. Also find the equivalent integral by reversing the order and evaluate it.
14. Define the concept multivariable function and their extreme values with example. Find the extreme value of the function $f(x,y) = 16 - (x+2)^2 - (y-2)^2$ if exist.

Group - C

Attempt any three questions. (Long Questions)

3×12 = 36

15. Discuss the different techniques of testing the convergence of an infinite series with example. To test the convergence of an infinite series $\sum n$, which test do you want to prefer and why? Using your prescribed method test the convergence.

16. Define Beta and Gamma function. Prove that

$\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta \, d\theta$. Using Gamma function evaluate $\int_0^1 x^6 \sqrt{1-x^2} \, dx$.

17. A thin plate covers the triangular region bounded by the x-axis and lines $x=1, y=2x$ in first quadrant. The plates density at point $P(x,y) = 6x+6y+1$, find plate's mass, centre of mass, first moment and second moments about co-ordinate axis.

18. With example justify the necessity of Lagrange's method of undetermined multiplier for finding the extreme values of multi-variable function. Using the suitable process find the extreme value of the function $f(x,y) = x^2+3xy-5y^2$ subject to constraint $2x+3y=6$.

Handwritten calculations for question 18:

$$L = x^2 + 3xy - 5y^2 + \lambda(2x + 3y - 6)$$

$$\frac{\partial L}{\partial x} = 2x + 3y - 2\lambda = 0$$

$$\frac{\partial L}{\partial y} = 3x - 10y + 3\lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = 2x + 3y - 6 = 0$$

$$\Rightarrow \begin{cases} 2x + 3y = 2\lambda \\ 3x - 10y = -3\lambda \\ 2x + 3y = 6 \end{cases}$$

$$\Rightarrow \begin{matrix} 2x + 3y = 2\lambda \\ 3x - 10y = -3\lambda \\ \hline 2x + 3y = 6 \end{matrix}$$

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FAR WESTERN UNIVERSITY
Semester End Examination-2078

English Grammar and Composition (ENG 111/CSIT.111)

Faculty: Science and Technology (B.Sc. General/CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Group - A

Attempt all questions

(Very Short Questions)

8×3 = 24

1. If you wear casual clothing to a meeting, what message do you give does it refer to?
2. Why should supertasters and nontasters pay close attention to the foods they eat?
3. How can people make real money on Second Life?
4. Why did Kelly decide to climb up Half Dome Mountain in Yosemite?
5. Why masdar is considered to be a unique city?
6. How could the use of social networking sites affect the human mind and behaviour?
7. How does someone become a successful athlete?
8. How can technology be used in our daily lives?

Full
Full

Attempt any five questions.

Group - B

(Short Questions)

5×8 = 40

9. What do you think the main reason that people use social networking sites?
10. What qualities do you think Beatrice has that helped her to be successful in her goals?
11. Would you like to live in a city with no cars? Why or why not?
12. Do you think there is too much advertising? Where would you like to see less or no advertising?
13. Why do you think we perceive activities such as mountain climbing to be riskier than they actually are?
14. Do you believe that things can make you happy? Have you ever received something that made you very happy?

P.T.O.

Group – C

Attempt any three questions.

(Long Questions)

3×12 = 36

15. Can a person's own life experiences make him or her a more helpful person? Explain, using examples from your own observations and experiences.
16. What positive and negative effects can technology have on people's lives? Is visiting friends more enjoyable than chatting online with friends? Why or why not?
17. Should athletes or sports teams accept money from corporate sponsors in order to be successful? Discuss one or more specific sports in your description.
18. Describe your favourite dish and explain if it has personal or cultural importance to you.

Bitter

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FAR WESTERN UNIVERSITY
Semester End Examination-2078
Information Technology Fundamentals (CSIT.112)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Group – A

Attempt all questions

(Very Short Questions)

8×3 = 24

1. How can you use IT in business and Industry?
2. Why computer is known as versatile machine?
3. Convert $(356)_{10}$ into $(?)_8$ 544
4. List the functions of CPU.
5. Explain the types of data file.
6. Write short notes on internet service providers (ISP)?
7. Explain the importance of HTML in web page designing.
8. Why system software is needed in computer?

diff purpose of test
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Group – B

Attempt any five questions.

(Short Questions)

5×8 = 40

9. Define hardcopy and softcopy output. Differentiate between impact and non-impact printers with example.
10. What is the objective of using operating system? How operating system performs process, memory and file management activities.
11. Define number system. Subtract $(11011)_2$ from $(1011)_2$ by using 1's and 2's complement method.
12. What is language processor? Describe different types of language processors with examples.
13. Define simplex, half duplex and full duplex channel of communications with examples.
14. What is meant by random access? Explain different variations of ROM.

Group – C

Attempt any three questions. (Long Questions)

3×12 = 36

15. What do you mean by generation of computer? Differentiate between third and fourth generation computer.
 16. What is Boolean algebra? Describe any five logic gates with truth tables and gate symbols.
 17. Why do we need computer network? Discuss different types of network topologies along with their merits and demerits.
 18. How are computers classified on the basis of their size? Explain with example.
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Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2021

Bachelor in Computer Applications
Course Title: C Programming
Code No: CAUS151
Semester: II

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.
Group B

Attempt any SIX questions.

[6×5 = 30]
[5]

2. Write a C program to generate the following output using loop.

```

1 0 1 0 1
 0 1 0 1
   1 0 1
    0 1
     1

```

3. How will you define pointer? Write a program that illustrate how pointer variable change the value of normal variable. **(5)** **[2+3]**

4. Write a C program to calculate sum of the natural number up to N number using recursive function. **(5)** **[5]**

5. Write a C program to calculate following expression: $5x\sqrt{y^2+5}$ **(5)** **[5]**

6. Write a C program to copy contents of student.txt file into another file called info.txt. **(5)** **[5]**

7. Why do you use DMA instead of array? Explain DMA with suitable example. **(4)** **[2+3]**

8. Write a C program to generate following output using graphics functions. **[5]**

BCA

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Write a C program to store the player name, runs scored, wickets taken of 20 cricketers using structure along with pointer then arrange the players in order of maximum wickets to minimum wickets they had taken.

10. Write a C program to generate following series using function:

$$Sum = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

11. Define SDLC. Explain software process models. **(6)** **[2+8]**

Far-Western University
Department of Humanities and Social Sciences
Mid-term Examination – 2079

Subject: English Grammar and Composition (CEHG 101)
Level: Undergraduate
Semester: First

Full Marks: 55
Time: 1Hr. 40 minutes

(Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks)

Group A

Answer any Four.

4×10=40

1. What are some ways you can make a good impression? Write at least five tips for facing an interview.
2. What makes food taste differently? Discuss three types of tasters.
3. Do you think that businesses that sponsor sports like car racing would be as successful without giving sponsorship money? Why or why not?
4. Define collocation with example. Complete each sentence with the correct adjective + preposition collocation from the list below.

Afraid of, interested in, sure about, due to, involved in.

- a. Parents whose children compete in sports are often afraid of injuries.
- b. The player's injury was due to overuse.
- c. Carlos was not sure about the meaning of the word, so he looked it up in the dictionary.
- d. Are you interested in going to the baseball game tonight? I have an extra ticket if you'd like to go.
- e. More children are involved in professional sports at a very young age today. My neighbor's daughter started playing soccer when she was four.

5. What does the second life mean? What are the benefits of living a second life?

Group B

Answer any One

1×15=15

6. Besides television, what kind of technology would be most difficult for you to live without? Why?
7. Why do people help or hurt others? Explain the concept of bystander effect in detail with example from the book you have read.